**HEADSTART: Forecasting health-system crises from published evidence when local outcome data are scarce**

## 1. Summary

At the start of a health crisis, many useful inputs are already there. Weather, air quality, calendar effects, wastewater and public surveillance data come from public feeds or established partners. What is missing is not this context. It is local, timely and reliable data on the outcome of interest for the episode that is starting. Past local records may be long, but they do not say how the new episode will affect that outcome. The model has to estimate this relationship early and update it as outcomes arrive.

HEADSTART asks whether published findings from other settings and past events can improve forecasts before enough outcomes from the current episode are available, and whether the method can detect when those findings mislead. Studies can inform plausible parameter values, explanatory variables and lags, model forms, and outcome definitions or warning thresholds. A prior is a distribution of plausible parameter values before it is updated with outcomes from the episode being forecast. It represents uncertainty rather than treating a published estimate as exact. HEADSTART compares two strategies. Both use the same context feeds, historical local records and outcomes already observed in the current episode. Only the evidence-informed strategy also uses information extracted from published studies to guide its parameter distributions and model choices.

The work has four linked parts. First, I will compare automated extraction with a manual benchmark and measure how extraction errors change the uncertainty of the model inputs. Second, I will build a model that forecasts the chosen outcome and expresses it as ordered levels of pressure on the health system, with the number of levels and their thresholds taken first from the evidence and fixed before any test. The outcome will be selected after reviewing the literature and the available data, using registered criteria, before any model performance is examined. Third, I will replay past events using only the information available at the time and compare the two strategies. Fourth, I will test whether differences in forecast quality would change operational decisions.

The tests take place in Geneva and cover three types of crisis with different causes. The main test uses respiratory epidemics, for which published evidence is comparatively rich. The second uses heat events. The third, in year four, uses legionellosis and a linked case-environment dataset to forecast case incidence. If published evidence improves forecasts, the project shows when and by how much. If it brings nothing or causes harm, the result says where it should not be used. For the demand tests that pass validation and that the services authorise, forecasts run on a daily dashboard in observation mode and are scored as outcomes arrive. The dashboard does not direct clinical care.

I trained as an econometrician and spent eleven years in quantitative finance on rare-event risk before a doctorate in biomedical sciences at Geneva. I conceived and built LiteRev with Aziza Merzouki and Olivia Keiser and extended it into LiteRev-Evidence, the prototype this project builds on. Through GESICA and GeoAI4EI (Horizon Europe) I work with Geneva's surveillance data and with European epidemic-intelligence partners. HEADSTART builds on this work to test a distinct question, in a programme I will lead.

## 2. Proposed research

### 2.1 Current state of research in the field

#### 2.1.1 Forecasts need outcome history as well as context data

Syndromic surveillance detects departures from expected baselines [1, 2], and ensembles of forecasting models perform strongly when outcome data are rich [3, 4]. Emergency dispatch records

move ahead of confirmed surveillance. The autumn wave of the 2009 A(H1N1) pandemic was seen eight days early in three European regions [5], long series of emergency calls track influenza-like illness [6], and a systematic review maps the field [7].

The usual problem is not a lack of context data. Weather, air quality, calendar effects, wastewater and some surveillance indicators are available in near real time and can be linked automatically to a forecasting model. What is missing at the start is data on the outcome itself. A new episode often differs from earlier ones in the population affected, the drivers and the severity, so the link between context and outcome has to be checked again as outcomes arrive.

Google Flu Trends shows why indirect indicators cannot replace this check. It overestimated influenza by more than a factor of two [8]. Published findings may help, provided their uncertainty and their relevance to the local setting are assessed.

### 2.1.2 Published studies can inform forecasts by assessing uncertainty and relevance

Statistics has methods for using earlier studies in a new setting while keeping their uncertainty and judging their relevance. These include power priors [9], commensurate priors [10] and robust meta-analytic-predictive priors [11]. Informative priors have also been used to forecast diseases with little local data [12]. What has not been shown is whether an automated pipeline can extract published evidence, keep its uncertainty, judge its relevance to the new setting and improve forecasts of local outcomes at the start of a crisis.

Published studies offer more than parameter values. They suggest which variables and time lags matter, which types of models work, and how to define outcomes and warning thresholds. In some fields this evidence is kept up to date in living systematic reviews [13]. To my knowledge, these different forms of evidence have never been combined and tested together in a real forecasting setting.

Automated extraction makes large-scale evidence synthesis possible, but numbers are extracted less reliably than categories (reported accuracy roughly 47–88% against 74–96%), and omissions make up a large share of the errors [14]. HEADSTART tests whether these errors understate uncertainty in a systematic way and whether a measurement-error model can correct them.

Even a correctly extracted estimate may not apply locally, because populations, case definitions, service organisation and policy differ. Formal methods to assess transferability exist [15–17], but they have not been combined with automated extraction, and their effect on forecasts at the start of a crisis has not been measured.

### 2.1.3 Forecast demand and levels of service pressure

Emergency services need to know not only how many calls or visits to expect, but whether demand is heading for a level that calls for action. Surveillance research has used hidden Markov models to separate epidemic from non-epidemic periods [18, 19], and extreme-value methods describe rare peaks in demand [20], including hospital congestion in Switzerland [21]. HEADSTART expresses its forecasts as ordered levels of pressure on the system, so that all candidate models can be compared on the same scale. The number of levels and their thresholds are taken first from the published definitions of outcomes and warning thresholds, and four levels (routine, elevated, strained, critical) are the default only where the literature is silent. Transmission models [22] are considered only for infectious crises, and heat is modelled through exposure and response. Rises in variance and in day-to-day correlation, which precede some transitions [23] and have been studied in epidemics [24–26], are tested as possible extra warning signs, not assumed to be reliable.

### 2.1.4 Forecast quality is important when it affects a decision

A forecast is useful when it changes a decision for a good reason. Proper scoring rules judge probability forecasts, including their calibration [27], and decision analysis judges them by their consequences [28]. Heat thresholds built on mortality need not match those for service demand [29], and the link between air pollution and asthma emergencies varies between places [30]. HEADSTART therefore scores forecasts first statistically, then against the costs of wrong decisions as stated by emergency-service experts.

### 2.1.5 The gap, evaluating evidence in a different context

Existing efforts include the ECDC RespiCast and CDC FluSight forecasting hubs, the GeoAI4EI project (Horizon Europe) and the GESICA project (France and Switzerland). HEADSTART asks a different question. When local outcome data are still few, does evidence from earlier studies improve automated forecasts of what a health service will face? It answers with a comparison registered in advance, the same data streams for both strategies, and a replay of the past that excludes later information.

## 2.2 Current state of personal research and competences required for the project

### 2.2.1 From econometrics and finance to public health

I studied quantitative economics and econometrics (MSc and DEA, University of Geneva), then worked for eleven years in quantitative finance on risk, extreme values, regime models and stress testing, and became a certified Financial Risk Manager (FRM). This is where my practice of rare-event forecasting, regime-switching models and Bayesian modelling comes from. In 2019 I joined the Institute of Global Health and completed a doctorate in biomedical sciences (defended 18 December 2023).

### 2.2.2 Doctorate, prediction under sparse information and LiteRev

In [31] I predicted individual HIV status from social and behavioural characteristics across East and Southern Africa and showed where the models transferred between countries, in a line of work shared with [32] and [33]. This work gave me direct experience of when a model learnt in one setting holds in another.

I conceived and built LiteRev [34] with Aziza Merzouki (PhD, computer science) and Prof. Olivia Keiser. The tool runs systematic searches from plain-language or Boolean queries, removes duplicates and maps the literature. LiteRev was funded by UNIGE (CHF 30,000), the Fondation Ernst et Lucie Schmidheiny (CHF 10,000), Venture Kick (CHF 10,000) and Mimosa (CHF 20,000). The GESICA systematic review of AI in emergency medical services [7], which I co-authored, uses LiteRev.

### 2.2.3 Outbreak and health-system modelling in Switzerland

In [35] I compared the clinical severity of Delta and Omicron sub-lineages in a Swiss university hospital. [36] built age-structured scenarios of the Swiss SARS-CoV-2 epidemic to support decisions under time pressure. I also contributed to WHO African Region reporting and to seroprevalence estimates [37]. As last author of [38] (accepted), I had the senior role in the statistical analysis of the classification of self-reported cardiovascular disease across harmonised WHO STEPS surveys.

### 2.2.4 Since 2024, LiteRev-Evidence, GESICA and GeoAI4EI

Since 2024 I have extended LiteRev into LiteRev-Evidence, a prototype for extracting numbers from studies and turning them into models. Today it searches and deduplicates the literature, records screening decisions with PRISMA accounting, extracts study details and PICO elements (population, intervention, comparator and outcome), keeps the source and a certainty grade for each number, and pools the numbers into distributions of plausible parameter values. It also suggests explanatory variables with their supporting studies, lists candidate outcomes, fits models of several families with

time-series validation, and builds compartmental model parts from published parameter ranges. Evidence updates are reviewed before they are applied. All of this exists as a prototype only. None of it has been verified yet. In particular, the model-selection pipeline still has to show that it uses only information available at each forecast date. That validation is the work of this grant.

GESICA is a large Franco-Swiss consortium. In it, the Geneva group led by Prof. Olivia Keiser works on infectious-disease crises, starting with influenza, COVID-19 and RSV. My contribution was to list the variables and data sources for these diseases, and for heat waves, in an inventory of 28 surveillance sources (23 on diseases and 5 on the environment), each described by coverage, resolution, latency, access and quality. HEADSTART uses this inventory. Within GeoAI4EI (Horizon Europe, 2026–2030) I contribute to the European epidemic-intelligence toolbox. The question of HEADSTART grew out of the meeting between LiteRev and these two projects on predicting health crises.

The prototype lowers the development risk, since search, extraction and model selection already run in a first form. The grant is about the open scientific questions, whether the extracted evidence is reliable enough, whether it transfers to the local setting, and whether it improves forecasts.

#### 2.2.5 The legionellosis study

I carry out the full analysis of a cantonal study already under way (BASEC 2026-00324, ethics granted, PI Prof. Olivia Keiser) that links confirmed legionellosis cases in Geneva to individual domestic hot-water installations. These linked data are the basis of the third test, on legionellosis case incidence, in year four. The dataset covers ten years of confirmed cases. Because cases are few, incidence is aggregated to monthly or, where counts allow, weekly series, and the horizon is set at registration.

Together these steps give me the methods for forecasting rare events, the evidence pipeline, and the Geneva and public-health context that HEADSTART needs.

### 2.3 Detailed research plan

#### 2.3.1 Objectives and hypotheses

The central hypothesis is that published evidence can improve forecasts early in an episode by helping estimate how the available context relates to the outcome, provided extraction uncertainty and relevance to the local setting are assessed.

| | Role | Statement |
|---|---|---|
| H1 | Evidence quality | Do automated methods preserve the uncertainty in published numbers? |
| C2 | Model adequacy | Can the model reliably detect and assign probabilities to different levels of pressure? |
| H3a | Main hypothesis | Does adding published evidence improve early forecasts when outcome data from the current episode are few? |
| H3b | Safety check | Can adaptive weighting reduce harm when published evidence conflicts with local outcomes? |
| H3c | Secondary question | Do short-term changes in variability add useful predictive information? |
| H4 | Decision value | Does any gain in forecast quality change an operational decision for the better? |

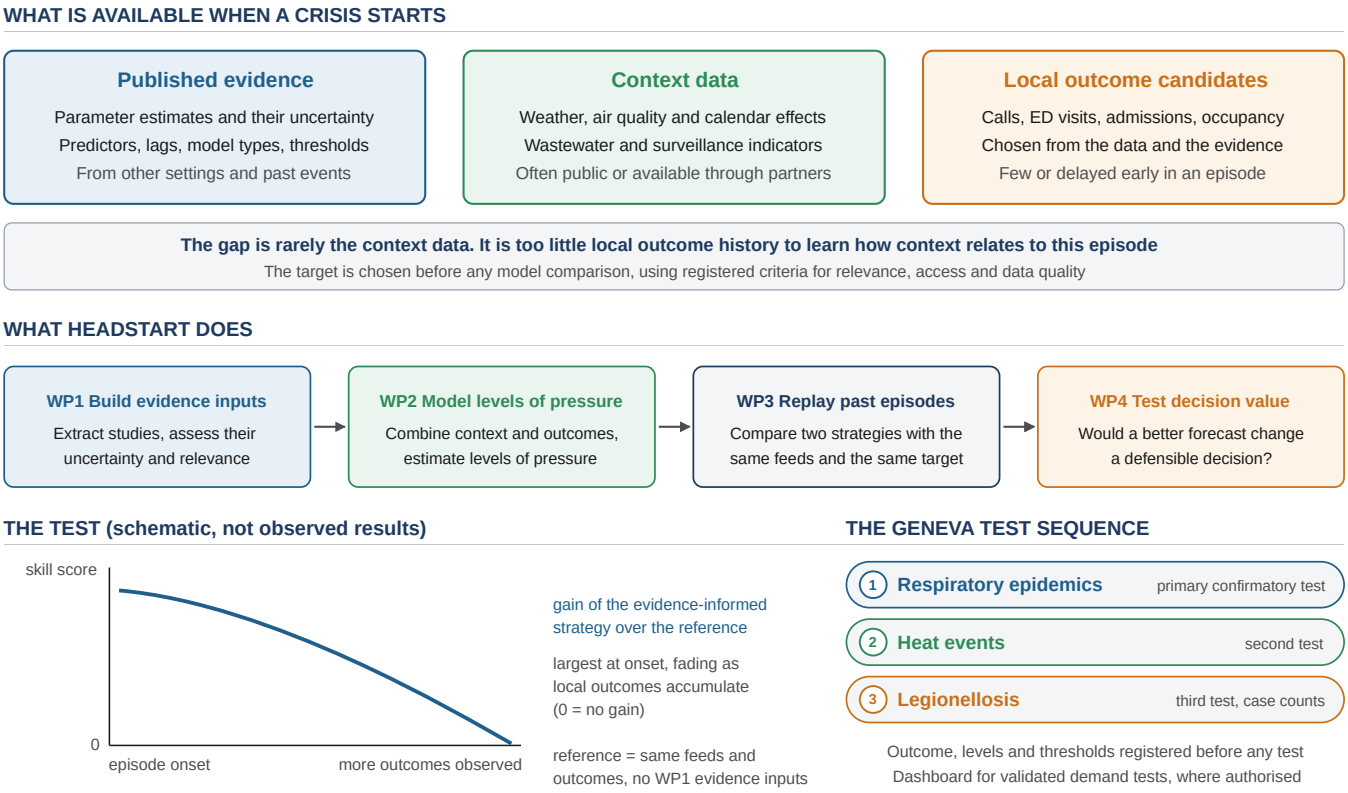



*Figure 1. What is available when a crisis starts, the gap, what HEADSTART does (WP1 to WP4), the test, and the Geneva test sequence.*

### *O1. Make published evidence usable while acknowledging its uncertainties*

H1. When numbers are extracted from studies automatically, part of the uncertainty that the studies report is lost, both the uncertainty inside each study and the differences between studies. The distributions of plausible parameter values then come out too narrow. Adding a correction for extraction error should widen them back to a realistic size. To test this, a sample of studies is extracted manually and used as the reference. The automated results, with and without the correction, are compared with this reference. The correction is validated if it brings the spread ratio of T1.3 close to one and improves interval coverage on the held-out studies. Studies used to tune the correction are kept out of this test set.

Automated extraction also misses numbers entirely [14]. H1 tests whether these omissions narrow the distributions further and whether the correction restores a useful level of uncertainty. The benchmark is built on parameter values. The other kinds of evidence (variables, model forms, thresholds) are compared with published reviews instead.

### *O2. Represent escalation in a form that separates the state from the point forecast*

C2. Model adequacy criterion. The model must give levels of pressure that make sense to the services and probabilities that are right on average, at the same false-alarm rate for every model. This common scale lets all candidate models be compared in operational terms. It does not assume that the regime-switching model beats simpler threshold methods.

Simulated data first test whether the levels and the moves between them can be recovered from the available data. Forecast calibration is the second check. If the model fails either check, the simpler ordinal model registered as fallback is used and the main comparison still goes ahead.

### *O3. Test whether published evidence improves forecasts and identify instances where it is detrimental*

H3a. In the first days of an episode, a strategy that uses published studies (for parameter values, explanatory variables, lags and model forms) will forecast better than a reference strategy that uses the same context data, local history and outcomes seen so far, but no model inputs derived from the published evidence in WP1. How long it lasts is secondary.

H3b. Adaptive weighting lowers the weight of the published evidence when local outcomes contradict it. When the evidence is right, adaptive weighting should do no worse than fixed weighting, within a margin Δ. When the evidence is deliberately shifted in location, scale or relevance to the local setting, adaptive weighting should do better. This checks that the model limits harm instead of trusting the literature blindly. Δ is half of the smallest improvement counted as meaningful under H3a, so adaptive weighting may give up at most half of that benefit when the evidence is right.

H3c. Two simple signals, the recent variability of demand and how strongly each day resembles the day before, are tested as extra predictors. The question is whether they add anything to the context data, the local level and trend, and the published evidence.

Primary comparison. Both strategies choose their models with the same procedure and are scored with the continuous ranked probability score (CRPS), which rewards forecasts that put high probability near what actually happened. The main test uses respiratory episodes in their first days. The heat test is always run, but it counts as confirmatory only if the respiratory test passes, which keeps the overall false-positive rate under control. A secondary analysis follows how the difference evolves as local outcomes come in.

### *O4. Establish whether predictive improvement matters for decision-making*

H4. The forecast with the best score is not always the most useful one for the services. The models are therefore also judged on the decisions they would trigger, using the costs of acting too early, too late or for nothing, as stated by emergency-service experts. An evidence-informed forecast counts as useful only if it changes a decision that matters in practice.

### *Scope. Three Geneva crisis types*

HEADSTART covers three Geneva crisis types. (1) Respiratory epidemics (SARS-CoV-2, influenza and RSV, together or one of them) are the main test because the published evidence is comparatively rich. (2) Heat events are the second test, with ozone and PM10 as co-exposures or effect modifiers rather than separate domains [39, 40]. (3) Legionellosis is the third test, in year four, on the linked case-environment data, and forecasts case incidence rather than demand. No fourth domain is planned. The three crisis types are fixed. Within each, the outcome measure, the population and, for respiratory epidemics, the pathogen are not fixed in this plan. They depend on what the published evidence defines and on the local service data available, for example emergency calls filtered on the recorded symptoms, emergency-department visits, hospital admissions, bed or ICU occupancy, or case counts. The candidates are listed at registration. The target is then chosen after an audit of access, completeness, cause coding, reporting delay, time resolution and usefulness for the services, before any forecast performance is compared, and it does not change afterwards. Open public series support the validation of the methods but are not presented as a substitute for local service data. A month-12 data gate checks that the selected target meets the registered criteria, otherwise the next candidate takes its place. The project is not intended to outperform established forecasting hubs when long outcome histories are available. It does not assume that published evidence will improve forecasts in every setting, and it does not promise clinical decision-making by an automated system. The dashboard, where authorised, will operate in observation mode only.

### 2.3.2 Work packages and methods

*WP1. Turn published evidence into model inputs (M1–M20)*

#### T1.1. Fix what to look for and register the protocol (M1–M4)

Before any extraction starts, a written protocol fixes what the project looks for in the literature. There are four kinds of evidence. (1) Numbers, the core of the work, such as how demand responds to temperature, heat days, ozone and PM10, how large and how fast surges are, how long patients stay, how full units get, and what share of patients is admitted. Transmission parameters and conversions from wastewater signals to cases come second. (2) The variables and time lags that studies report as useful for each crisis type. (3) Which model families work for which outcome and horizon [41]. (4) How studies define outcomes and warning thresholds [29]. The numbers are the subject of the extraction benchmark. The other three kinds are recorded with their source and a certainty grade and compared with published reviews.

#### T1.2. Build the manual benchmark (M3–M9)

For each crisis type, a systematic review run under the registered protocol identifies the relevant publications, and these form the benchmark. A second extractor, hired for this task, and I each extract the target numbers by hand from all of them when the set is small, and otherwise from a random sample drawn to cover the different kinds of parameters, reporting quality, study designs and source types, including journal articles and dated situation reports. The sample size is fixed at registration from the size of the set. The manual benchmark covers the parameters that the main models actually use, not every number reported in each field, and the secondary parameter classes enter only if the main models need them. Differences are reconciled against the source text with a written coding guide. Values that stay ambiguous are flagged and left out of the main reference set, and their effect is checked in a sensitivity analysis. Error can then be measured separately for the sources used only in secondary analyses. As an outside check, the automated search is compared with the study lists of published systematic reviews, and the pooled numbers with the summary estimates those reviews report. The benchmark is published openly.

#### T1.3. Measure the errors of automated extraction (M7–M14)

The automated extraction is compared with the manual benchmark on four things, the values themselves, their uncertainty, the numbers it missed, and how much studies differ from each other. The main measure for H1 is simple. For each kind of parameter, the spread of the automated distribution is divided by the spread of the manual one, and the typical ratio is reported. A ratio below one means the automated extraction made the distribution too narrow. Uncertainty inside studies and differences between studies are tracked separately and together, and the analysis is repeated with each version of the extraction model.

#### T1.4. Turn the numbers into distributions with their uncertainty (M12–M18)

Extraction errors are treated as measurement error and carried into the pooled distributions. Several ways of pooling studies are compared, weighting by precision, weighting by study quality, meta-analytic-predictive priors and power priors. Each resulting distribution keeps a record of its sources, the extraction uncertainty and the weight given to the outside evidence.

#### T1.5. Check how well each finding transfers (M15–M20)

Each finding is screened for how well it transfers to the local setting, looking at the factors that change an effect and at the differences between the study setting and the local one. The screening criteria,

and how a low score widens the distribution or lowers the weight of the evidence, are fixed before the forecast comparison starts.

Deliverables are D1.1 (open extraction benchmark), D1.2 (extraction-error analysis) and D1.3 (evidence-derived model-input library).

Risk and mitigation. If manual extraction time is limited, the benchmark narrows to the core parameters.

### *WP2. Model the levels of pressure on the system (M1–M28)*

#### T2.1. Build and test the pressure model (M1–M9)

The model follows how pressure on the system moves between ordered levels, for example routine, elevated, strained and critical, and gives the probability of each level for each day. It is a Bayesian regime-switching model [42]. Its main input is the registered outcome, with other outcomes such as emergency-department visits and ICU occupancy added when access allows. Weather, air quality, calendar effects and epidemic indicators are the explanatory variables. The model is first tested on simulated data to check that it can find the levels and the moves between them, also with a single data series. If it cannot, a simpler ordinal model is used. The number of levels and their thresholds are fixed at registration, from the literature where it gives usable definitions. Otherwise four default levels are set on seasonal percentiles (routine below the 75th, elevated to the 90th, strained to the 97.5th or when capacity is under sustained pressure, with ICU occupancy above 85% or emergency-department boarding above its seasonal 90th percentile, and critical above the 97.5th or when capacity is saturated). The highest level that applies counts. These thresholds guide the model and are not hard cut-offs. Experts can later refine what each level means for decisions, not the definitions used in the main test.

#### T2.2. Model the rare peaks of demand (M6–M14)

Rare, very high demand is modelled with extreme-value methods (peaks over a threshold, generalised Pareto distribution) and linked to the probability of the critical level. The sensitivity to the chosen threshold is reported. It supports the main comparison and carries no hypothesis of its own.

#### T2.3. Bring the published evidence into the model (M10–M20)

The distributions built in WP1 feed the model parameters for which published evidence exists, such as weather effects, surge size and the moves between levels. Three ways of using them are compared. The first uses almost no outside information. The second uses the evidence distributions as they are. The third lowers the weight of the evidence when a registered conflict measure shows that local outcomes disagree with it. The conflict measure and the weight rule are fixed before evaluation. Power priors are run as a sensitivity check.

#### T2.4. Test two early-warning signals (M12–M20)

The two signals of H3c, the recent variability of demand and how much each day resembles the day before, are tested as optional predictors of a move to a higher level. The question is whether they add anything to the context data, the recent level and trend, and the published evidence. A negative answer is also a useful result.

#### T2.5. Calibrate, automate and check validity (M20–M28)

Forecast probabilities are calibrated with methods made for data that change over time. Public feeds (weather, air quality, wastewater, surveillance indicators) are loaded automatically as often as they are updated. Partner data are linked only under access agreements. The pipeline records when each value was observed and released, which version of the data and of the model was used, refits the models on

a fixed schedule and checks after each fit that the residuals, the spread and the calibration are acceptable. Models that fail these checks are set aside. The working code is released within LiteRev-Evidence.

Simulated data are used only to test the model, never as extra observations. Treating simulations based on the literature as real outcomes would count the same information twice and spoil the test of conflict between the evidence and the local data.

Deliverables are D2.1 (model specification and identifiability study), D2.2 (open implementation) and D2.3 (methods paper).

Risk and mitigation. If the states cannot be separated reliably, the registered ordinal model is used and the identifiability limit is reported.

### *WP3. Replay past crises and test the forecasts (M1–M42)*

#### T3.0. Choose the outcome, check the data, register the rules (M1–M14)

Outcomes. The outcome of each test is not fixed in advance. It depends on what the published evidence defines and on the data available, for example emergency calls (CASU-144) filtered on the symptoms and urgency recorded for each call (audio recordings are not used), emergency-department visits by cause, hospital admissions, bed or ICU occupancy. The candidates are listed at registration in order of preference. Provisionally, respiratory epidemics use CASU-144 calls by symptom first, then HUG emergency-department visits, then admissions or occupancy. Heat events use the same service series restricted to heat-related causes, then a hospital outcome. Legionellosis uses weekly or monthly case counts. The final order is registered after the data audit. The target is chosen on access, completeness, cause coding, reporting delay and revisions, time resolution and usefulness for the services, before any forecast is compared. The causes and population groups used to filter it come from the evidence. Once chosen, the target stays fixed, so that no outcome can be picked afterwards because it gives better results, and both strategies are scored on the same target, period and rules. The other candidates serve as extra inputs for the pressure model. Weather, air quality, wastewater, sentinel consultations and calendar effects are context inputs. The design is single-canton on purpose, and transfer is tested across the three crisis types.

Data gate. At month 12 the chosen target is checked against criteria on history length, resolution, reporting delay and completeness fixed in advance. If it fails, the next outcome in the list takes its place. Performance plays no part in this choice.

Episode eligibility. An episode enters the main comparison only if its start can be identified without later information, it has enough history before it and enough outcomes after it, the literature available at each forecast date can be rebuilt, and it can be separated from neighbouring episodes. Other episodes serve for sensitivity analyses. Co-circulating respiratory viruses count as one episode.

Registration. The outcome list, the gate and the eligibility rules are registered first. After the month-12 check and a blinded inventory of the eligible episodes, but before any performance is seen, a second registration fixes the early-episode window N, the main forecast horizon per crisis type and the margin Δ. Horizons follow what the services need, not results.

Data hierarchy. The main test uses local service data, chosen from the registered list. If a series is not available in time or fails the gate, the next one in the list takes its place. If no local series can be obtained, the pipeline is run on open Swiss series (federal respiratory surveillance, national wastewater data, weekly deaths and MeteoSwiss weather), with archived multi-model forecasts as comparators [3, 4], to show that the methods work and to release a public benchmark, and the claim is

narrowed to that. The HUG COVID-19 hospitalisation series I analysed before [35] can serve as an additional check if it can be obtained again.

### T3.1. Assemble the data (M12–M20)

The outcome series are lined up with the context feeds and with each other on the grid of the chosen outcome, daily for the demand tests. Missing values and reporting delays are measured, and recent days that are still incomplete are corrected with nowcasting [43, 44], because reporting tends to slow down under pressure and can look like an early warning. The HUG COVID-19 hospitalisation data I analysed before [35] may be requested again as an extra check. Pharmacy sales and wastewater measurements are requested through the cantonal pharmacist and used as early indicators.

### T3.2. Replay with only what was known at the time (M18–M30)

For each past episode, forecasts are made at successive dates using only what was known on that date. The evidence comes from papers and preprints published before each date. Dated WHO and ECDC reports about events elsewhere are used only in a secondary analysis that tests a wider evidence feed. Any text describing the local episode itself is excluded, because it could give away the outcome. Every input carries two dates, when it was observed and when it became available. When a source was later corrected, the replay uses the version available at the time, not the corrected one. Automatic feeds are replayed at their own frequency. The candidate horizons are 7, 14 and 28 days, and the main horizon is fixed at the second registration on what the services need. The early window of each episode starts at a rule that uses only information known at the time.

### T3.3. Build the model library and run the tests (M20–M34)

Two strategies are registered before evaluation. The reference strategy uses a registered set of standard forecasting models (seasonal regression, persistence, surveillance alarm methods [45], machine-learning learners and the WP2 pressure model with almost no outside information), fitted to the local history, the outcomes already seen in the episode and the context feeds, with no model inputs derived from the published evidence in WP1. The evidence-informed strategy uses the same inputs plus the WP1 distributions and the registered advice on variables, lags and model forms, in three versions, fixed evidence weights, adaptive weights, and adaptive weights plus the two signals of H3c. Transmission models are allowed only for infectious crises. The allowed candidates are fixed per crisis type at registration.

At each forecast date, each strategy picks its best candidate using only outcomes seen before that date, with the CRPS as the ranking criterion and log score and calibration as entry checks. The chosen model then forecasts the following period, whose outcomes played no part in the choice. Both strategies follow the same rules. The main comparison tests the whole evidence-informed strategy, and secondary comparisons separate the effect of the priors, of the variable and lag advice, and of the model forms.

The tests run in a fixed order. Test 1 compares the two strategies on the CRPS skill score over the early window of respiratory episodes, two-sided at $\alpha = 0.05$, because harm matters as much as benefit. Success needs a paired permutation p-value below $\alpha$ and a lower confidence bound above the registered minimum improvement. Test 2 repeats the comparison for heat. It is confirmatory if Test 1 passes and exploratory otherwise, and if Test 1 fails H3a is not supported. Whatever the result, the effect sizes and the conditions under which the evidence helped or harmed are reported. The reference models and their settings are fixed at registration so that the comparator cannot be weakened later.

Main endpoint. For each episode and strategy, the CRPS is averaged over the forecast dates in the window at the main horizon, and the skill score is one minus the ratio of the two averages, so that

positive values favour the evidence-informed strategy. The main statistic is the mean over episodes. The episode is the unit of analysis. The number of eligible episodes is established with the registered eligibility rules before any forecast performance is examined, and it is expected to be small. A paired permutation test and a bootstrap over whole episodes are both registered. Simulations run before the second registration set the type-I error, the power, the smallest number of episodes for a confirmatory claim and the window N, without using any forecast result, and check that the ratio-based skill score stays stable when the reference CRPS differs between episodes. If fewer eligible episodes exist than that minimum, the analysis is reported as an estimate with its uncertainty, not as a confirmatory test. The minimum relevant improvement is fixed at the second registration by a rule based on operational relevance, and Δ is half of it. Secondary measures are the log score, calibration and the detection of escalation at matched false-alarm rates. For H3b, the right-evidence case is a simulation, registered in advance, in which the evidence given to the model and the process that produces the outcomes agree within a set tolerance. The score compares adaptive weighting with fixed weighting, and positive values mean adaptive is better. Adaptive weighting passes if the score stays above minus Δ when the evidence is right, and is positive when the evidence has been shifted on purpose.

#### T3.4. Map when the evidence helps and when it harms (M28–M38)

The analysis finds the episodes where the published evidence helped and those where it hurt, and looks for the reasons, such as differences in population, outcome definition, service organisation, policy or period, extraction uncertainty, or conflict with the local outcomes. Deliberately shifted evidence serves as a stress test. A key question is whether adaptive weighting reduces harm early enough. The result is a map of when the approach helps, when it does not, and why, not a single average.

#### T3.5. Heat and legionellosis tests (M34–M42)

The heat test is the second test. The legionellosis test, in year four, applies the same comparison to case-incidence forecasts at the start of an outbreak, using the linked case-environment data. It uses the weekly or monthly case series, a horizon matched to that frequency, and its own rolling evaluation, outside the episode-level comparison of the two demand tests.

Deliverables are D3.1 (reproducible historical forecast-evaluation pipeline), D3.2 (primary result), D3.3 (benefit and failure analysis) and D3.4 (analysis across crisis types).

Risk and mitigation. Access to operational records may be delayed. Agreements are being pursued before the award. If no local series can be obtained, open Swiss data show that the methods work, but they cannot carry the main claim, which is then narrowed.

### *WP4. From forecast skill to decision value (M24–M48)*

#### T4.1. Elicit costs and thresholds with the services (M24–M32)

A structured elicitation (SHELF method) with two to five experienced people from HUG emergency medicine, CASU-144 regulation and capacity management puts numbers on the cost of escalating too early, too late or for nothing. The decision thresholds come from these costs. The exercise sets transparent thresholds and is not a representative survey. The decision studied is fixed in advance, namely escalating surge capacity when the forecast probability of the upper levels (strained or critical in the default scheme) passes the threshold. Models that forecast counts rather than levels are converted to the same level probabilities by a registered rule.

#### T4.2. Judge the forecasts by the decisions they trigger (M30–M40)

The WP3 forecasts are re-assessed on the decisions they would trigger, with net-benefit and decision-curve methods, to see whether the ranking of the models changes once the costs of wrong decisions

are counted. Calibration and threshold performance are also checked across groups by age and sex. A model that is well calibrated overall but poor for a group that matters for decisions does not pass.

#### T4.3. Dashboard and living updates (M34–M48)

For the demand tests that pass the forecast and decision checks and that the services authorise, the project runs a daily dashboard in observation mode. It publishes forecasts, level probabilities and validity checks, records its outputs and scores them as outcomes arrive, and it does not direct clinical care. New studies are checked daily, and full syntheses are refreshed on a registered schedule [13]. Any change to priors, variables, thresholds or model choice is version-controlled and applied only under pre-registered rules. Without authorisation, the retrospective evaluation and the update protocol remain the deliverables.

Deliverables are D4.1 (cost structure and fairness audit), D4.2 (decision-analytic evaluation) and D4.3 (the update protocol and, where authorised, the dashboard).

Reproducibility and data protection. Code is version-controlled in R and Python under a registered analysis plan. Clinical data stay in the UNIGE secure environment under the applicable approvals. The benchmark and the software are released openly, with synthetic data where real data cannot be shared.

The project delivers peer-reviewed papers and reusable resources (the extraction benchmark, the library of evidence-derived inputs, the decision-analysis framework and a public forecasting benchmark where open data are used), and reports its results whether the evidence helps, has no effect or causes harm.

### 2.3.3 Research environment, team and resources

The project is based at the Institute of Global Health (ISG), Faculty of Medicine, University of Geneva, which brings together epidemiology, infectious-disease modelling and evidence synthesis. The programme is led independently alongside the Institute's groups.

| Entity | Person in charge | Role in HEADSTART | Status |
|---|---|---|---|
| ISG, UNIGE | Prof. Olivia Keiser, contact person | Host institute | Hosting agreed, SNSF letters in preparation |
| DS4DH, UNIGE | Prof. Douglas Teodoro | Biomedical text mining for WP1 | Letter of support signed, re-issued for the HEADSTART title |
| HUG, Service des urgences | Prof. Thibaut Desmettre, médecin-chef de service | ED and ICU data, clinical partner | Extraction in discussion, letter requested |
| Unité urgences santé (144), HUG | Dr Robert Larribau, head of unit | Emergency-call data, clinical partner | Access held within GESICA, project agreement in preparation |
| GESICA (Franco-Swiss) | Prof. Olivia Keiser (UNIGE PI), Prof. Douglas Teodoro (Swiss data inventory) | Surveillance inventory, data access, emergency-medicine partners | Running |

| Entity | Person in charge | Role in HEADSTART | Status |
|---|---|---|---|
| GeoAI4EI (Horizon Europe) | Prof. Olivia Keiser, co-PI | European epidemic-intelligence partners, exchange of methods | Running 2026–2030 |
| Legionellosis study (data via the médecin cantonal and SIG) | Prof. Olivia Keiser, study PI | Linked case-environment data, analysis by the applicant | Study approved, BASEC 2026-00324 |
| HUG COVID-19 hospitalisation series | | Secondary validation and calibration series [35] | Re-request planned, subject to availability |
| Pharmacien cantonal | | Wastewater and pharmacy sales | Contacts established |
| UNIGE HPC and secure data environment | | Computing | In place |
| Open public series | | Surveillance, wastewater, deaths, weather | Public |

The ED and ICU extraction and the project-specific CASU-144 agreement are not yet secured. Each rests on an existing working relationship.

Project funds are capped at CHF 250,000 over four years, and my salary is separate. The budget entered in mySNF gives the detailed cost calculation.

| Item | Rationale |
|---|---|
| Research assistants (master's students, at most 40%) | Second extraction arm of the T1.2 benchmark (M3–M9), data handling, evaluation pipeline, workshop organisation |
| Database and dashboard services | Set-up of the project database, secure-environment runs and, where authorised, dashboard hosting |
| Conferences, two to three per year, and workshops | Presentation of results, exchange with the forecasting hubs and the GESICA and GeoAI4EI partners |
| Other eligible direct costs | As justified in the final SNSF budget |

Before month 1 and outside the grant, I will finalise the data agreements with HUG and CASU-144, confirm data reuse under the applicable governance rules, and harden LiteRev-Evidence before clinical data are connected.

## 2.4 Schedule and milestones

I lead all four work packages. Research assistants (master's students) provide the second extraction arm from months 3 to 9 and support data handling and the evaluation pipeline. The work packages run in parallel with fallbacks. WP2 can use weakly informative distributions if extraction error proves too large, and the retrospective decision analysis of WP4 does not depend on live deployment. Open Swiss data are the fallback if no local series can be obtained, so access agreements remain a priority before the award.

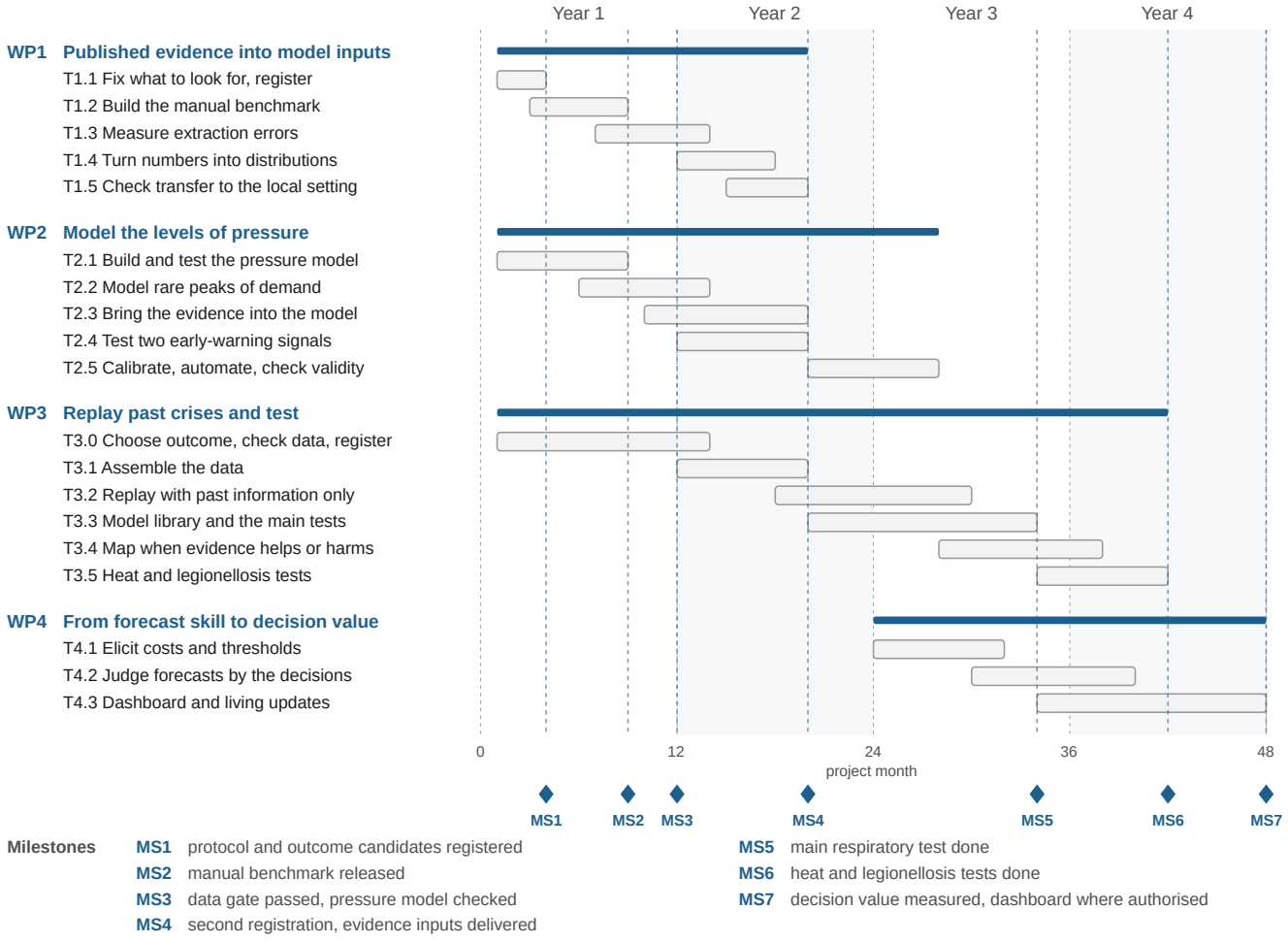



*Figure 2. Work packages, task durations and milestones across the 48-month project.*

| Milestone | Month | Criterion |
|---|---|---|
| MS1 | 4 | Extraction protocol and outcome candidates registered (first registration) |
| MS2 | 9 | Manual benchmark released |
| MS3 | 12 | Data gate passed, pressure model passes its checks (or fallbacks chosen) |
| MS4 | 20 | Second registration done, evidence inputs delivered |
| MS5 | 34 | Main respiratory test done |
| MS6 | 42 | Heat and legionellosis tests done |
| MS7 | 48 | Decision value measured, dashboard for the validated demand tests where authorised, all results reported |

## 2.5 Relevance and impact

If published evidence does not improve forecasts, that is still a useful result. It shows where using published estimates as model inputs adds little or carries too much risk, and spares modellers and evidence platforms an untested assumption.

Results are published in open-access journals, with preprints where appropriate. The extraction benchmark, the library of evidence-derived inputs, the code and, where open data are used, the forecasting benchmark are released as versioned, citable resources, and the findings go back to the participating services. The results and tools are also shared with GESICA and GeoAI4EI, which can carry them to the Franco-Swiss and European epidemic-intelligence networks.

The project will show whether outside evidence improves forecasts in the first weeks of a crisis, under which conditions, and with how much certainty. For the demand tests that pass validation and are authorised, it delivers a registered forecasting configuration and a dashboard in observation mode. Wider deployment to other crises is a possible next step after the grant, not a promised outcome.

### 2.6 Relevance for personal career development

My academic path has moved from quantitative economics and financial risk modelling to biomedical research, prediction across settings and collaborative software development. I built the methods, the prototype and the data relationships that make this project feasible, mostly inside existing programmes. HEADSTART gives me my own scientific question and a programme that I lead.

| Programme | Objective | My role | Distinction from Ambizione |
|---|---|---|---|
| GESICA | Franco-Swiss crisis intelligence | Contributor, data sourcing for respiratory diseases | Decision support, no test of evidence-derived priors |
| GeoAI4EI (Horizon Europe) | European epidemic-intelligence toolbox | Contributor | Infrastructure, no test of borrowed evidence |
| Legionellosis study (ongoing) | Geneva waterborne outbreak determinants | Full analysis | Supplies the data of the third test |
| Ambizione | When published evidence improves forecasts, and when it misleads | PI, full-time | - |

GESICA and GeoAI4EI build epidemic-intelligence and decision-support tools. HEADSTART tests one thing they do not, whether published evidence is a sound input to forecasts.

LiteRev-Evidence can already turn published estimates into parameter distributions, but nobody has shown whether using them improves forecasts in a new setting. That question, the design that answers it and the confirmatory analyses are mine.

Prof. Olivia Keiser, my contact person at the Institute, co-built LiteRev with me, leads the legionellosis study and is my PI in GESICA (respiratory diseases) and co-PI of GeoAI4EI (Horizon Europe). We also work together on the MAS in Public Health, where I have taught statistics and epidemiology since 2024. Prof. Douglas Teodoro (DS4DH) leads the data inventory of the Swiss part of GESICA and supports the text-mining work of WP1. HEADSTART is my own programme, hosted at the Institute of Global Health and led by me, with research assistants on defined technical tasks. By month 48, a validated answer and a first-author record will be the basis for leading a group or a professorship.

## Bibliography


1. Farrington CP, Andrews NJ, Beale AD, Catchpole MA. A statistical algorithm for the early detection of outbreaks of infectious disease. *Journal of the Royal Statistical Society Series A* 1996;159(3):547–563.
2. Noufaily A, Enki DG, Farrington P, Garthwaite P, Andrews N, Charlett A. An improved algorithm for outbreak detection in multiple surveillance systems. *Statistics in Medicine* 2013;32(7):1206–1222. doi:10.1002/sim.5595
3. Cramer EY, Ray EL, Lopez VK, et al. (consortium, more than 50 authors). Evaluation of individual and ensemble probabilistic forecasts of COVID-19 mortality in the United States. *Proceedings of the National Academy of Sciences* 2022;119(15):e2113561119. doi:10.1073/pnas.2113561119
4. Sherratt K, Gruson H, Grah R, et al. (consortium, more than 50 authors). Predictive performance of multi-model ensemble forecasts of COVID-19 across European nations. *eLife* 2023;12:e81916. doi:10.7554/eLife.81916
5. Rosenkötter N, Ziemann A, Garcia-Castrillo Riesgo L, Gillet JB, Vergeiner G, Krafft T, Brand H. Validity and timeliness of syndromic influenza surveillance during the autumn/winter wave of A(H1N1) influenza 2009: results of emergency medical dispatch, ambulance and emergency department data from three European regions. *BMC Public Health* 2013;13:905. doi:10.1186/1471-2458-13-905
6. Bonora R, Ticozzi EM, Pregliasco FE, Pagliosa A, Bodina A, Cereda D, Perotti G, Lombardo M, Stirparo G. Telephone calls to emergency medical service as a tool to predict influenza-like illness: a 10-year study. *Public Health* 2025;238:239–244. doi:10.1016/j.puhe.2024.12.021
7. Edjinedja KL, Larribau R, Orel E, Cossus J, Elfahim O, Yendouname K, Vaussenat F, Teodoro D, Robert-Nicoud S, Barakat O, Desmettre T. Artificial intelligence in emergency medical services for disasters and health emergencies: a systematic review. Submitted, 2026.
8. Lazer D, Kennedy R, King G, Vespignani A. The parable of Google Flu: traps in big data analysis. *Science* 2014;343(6176):1203–1205. doi:10.1126/science.1248506
9. Ibrahim JG, Chen M-H. Power prior distributions for regression models. *Statistical Science* 2000;15(1):46–60.
10. Hobbs BP, Carlin BP, Mandrekar SJ, Sargent DJ. Hierarchical commensurate and power prior models for adaptive incorporation of historical information in clinical trials. *Biometrics* 2011;67(3):1047–1056.
11. Schmidli H, Gsteiger S, Roychoudhury S, O'Hagan A, Spiegelhalter D, Neuenschwander B. Robust meta-analytic-predictive priors in clinical trials with historical control information. *Biometrics* 2014;70(4):1023–1032. doi:10.1111/biom.12242
12. Cook JD, Williams DM, Walsh DP, Hefley TJ. Bayesian forecasting of disease spread with little or no local data. *Scientific Reports* 2023;13:8137. doi:10.1038/s41598-023-35177-6
13. Elliott JH, Turner T, Clavisi O, Thomas J, Higgins JPT, Mavergames C, Gruen RL. Living systematic reviews: an emerging opportunity to narrow the evidence-practice gap. *PLoS Medicine* 2014;11(2):e1001603. doi:10.1371/journal.pmed.1001603
14. Shankar R, Lim A, Qian X. Performance of large language models in data extraction for evidence synthesis: a systematic review. *Journal of Biomedical Informatics* 2026;181:105086. doi:10.1016/j.jbi.2026.105086

15. Bareinboim E, Pearl J. Causal inference and the data-fusion problem. *Proceedings of the National Academy of Sciences* 2016;113(27):7345–7352. doi:10.1073/pnas.1510507113
16. Dahabreh IJ, Petito LC, Robertson SE, Hernán MA, Steingrimsson JA. Toward causally interpretable meta-analysis: transporting inferences from multiple randomized trials to a new target population. *Epidemiology* 2020;31(3):334–344. doi:10.1097/EDE.0000000000001177
17. Degtiar I, Rose S. A review of generalizability and transportability. *Annual Review of Statistics and Its Application* 2023;10:501–524.
18. Le Strat Y, Carrat F. Monitoring epidemiologic surveillance data using hidden Markov models. *Statistics in Medicine* 1999;18(24):3463–3478.
19. Watkins RE, Eagleson S, Veenendaal B, Wright G, Plant AJ. Disease surveillance using a hidden Markov model. *BMC Medical Informatics and Decision Making* 2009;9:39. doi:10.1186/1472-6947-9-39
20. Coles S. *An Introduction to Statistical Modeling of Extreme Values.* Springer, 2001. doi:10.1007/978-1-4471-3675-0
21. Ranjbar S, Cantoni E, Chavez-Demoulin V, Marra G, Radice R, Jaton-Ogay K. Modelling the extremes of seasonal viruses and hospital congestion: the example of flu in a Swiss hospital. *Journal of the Royal Statistical Society Series C (Applied Statistics)* 2022;71(4):884–905. doi:10.1111/rssc.12559
22. Keeling MJ, Rohani P. *Modeling Infectious Diseases in Humans and Animals.* Princeton University Press, 2008.
23. Scheffer M, Bascompte J, Brock WA, Brovkin V, Carpenter SR, Dakos V, Held H, van Nes EH, Rietkerk M, Sugihara G. Early-warning signals for critical transitions. *Nature* 2009;461:53–59. doi:10.1038/nature08227
24. O'Regan SM, Drake JM. Theory of early warning signals of disease emergence and leading indicators of elimination. *Theoretical Ecology* 2013;6:333–357.
25. Brett TS, O'Dea EB, Marty É, Miller PB, Park AW, Drake JM, Rohani P. Anticipating epidemic transitions with imperfect data. *PLoS Computational Biology* 2018;14(6):e1006204. doi:10.1371/journal.pcbi.1006204
26. Southall E, Brett TS, Tildesley MJ, Dyson L. Early warning signals of infectious disease transitions: a review. *Journal of the Royal Society Interface* 2021;18(182):20210555. doi:10.1098/rsif.2021.0555
27. Gneiting T, Raftery AE. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association* 2007;102(477):359–378.
28. Vickers AJ, Elkin EB. Decision curve analysis: a novel method for evaluating prediction models. *Medical Decision Making* 2006;26(6):565–574.
29. Lung S-CC, Yeh J-CJ, Hwang J-S. Selecting thresholds of heat-warning systems with substantial enhancement of essential population health outcomes for facilitating implementation. *International Journal of Environmental Research and Public Health* 2021;18(18):9506. doi:10.3390/ijerph18189506
30. Zheng XY, Ding H, Jiang LN, Chen SW, Zheng JP, Qiu M, Zhou YX, Chen Q, Guan WJ. Association between air pollutants and asthma emergency room visits and hospital admissions in time series studies: a systematic review and meta-analysis. *PLoS ONE* 2015;10(9):e0138146. doi:10.1371/journal.pone.0138146

31. Orel E, Esra R, Estill J, Thiabaud A, Marchand-Maillet S, Merzouki A, Keiser O. Prediction of HIV status based on socio-behavioural characteristics in East and Southern Africa. *PLoS ONE* 2022;17(3):e0264429. doi:10.1371/journal.pone.0264429
32. Merzouki A, Estill J, Orel E, Tal K, Keiser O. Clusters of sub-Saharan African countries based on sociobehavioural characteristics and associated HIV incidence. *PeerJ* 2021;9:e10660. doi:10.7717/peerj.10660
33. Esra R, Carstens J, Le Roux S, Mabuto T, Eisenstein M, Keiser O, Orel E, Merzouki A, De Voux L, Maskew M, Sharpey-Schafer K. Validation and improvement of a machine learning model to predict interruptions in antiretroviral treatment in South Africa. *Journal of Acquired Immune Deficiency Syndromes* 2023;92(1):42–49. doi:10.1097/QAI.0000000000003108
34. Orel E, Ciglenecki I, Thiabaud A, Temerev A, Calmy A, Keiser O, Merzouki A. An automated literature review tool (LiteRev) for streamlining and accelerating research using natural language processing and machine learning. *Journal of Medical Internet Research* 2023;25:e39736. doi:10.2196/39736
35. Orel E, Sobel J, Zanella MC, Iten A, Kaiser L, Keiser O, Vetter P. Clinical severity of Delta and 3 different Omicron sublineages in patients hospitalized because of COVID-19 in a Swiss tertiary center. *CMI Communications* 2024;1(2):100005. doi:10.1016/j.cmicom.2024.100005
36. Estill J, Venkova-Marchevska P, Roelens M, Orel E, Temerev A, Flahault A, Keiser O. Future scenarios for the SARS-CoV-2 epidemic in Switzerland: an age-structured model. *F1000Research* 2020;9:646. doi:10.12688/f1000research.24497.2
37. Nwosu KD, Fokam J, Wanda F, Mama L, Orel E, Ray N, Meke J, Tassegning A, Takou D, Mimbe E, Stoll B, Guillebert J, Comte E, Keiser O, Ciaffi L. SARS-CoV-2 antibody seroprevalence and associated risk factors in an urban district in Cameroon. *Nature Communications* 2021;12:5851. doi:10.1038/s41467-021-25946-0
38. Ng'ambi W, Estill J, Keiser O, Merzouki FA, Orel E. Machine learning-based classification of self-reported cardiovascular disease history in Africa using harmonised multi-country WHO STEPS surveys: 2014–2019. Accepted for publication.
39. Schulte F, Röösli M, Ragettli MS. Risk, attributable fraction and attributable number of cause-specific heat-related emergency hospital admissions in Switzerland. *International Journal of Public Health* 2024;69:1607349. doi:10.3389/ijph.2024.1607349
40. Ragettli MS, Vicedo-Cabrera AM, Schindler C, Röösli M. Impact of the warm summer 2015 on emergency hospital admissions in Switzerland. *Environmental Health* 2019;18:66. doi:10.1186/s12940-019-0507-1
41. Wargon M, Guidet B, Hoang TD, Hejblum G. A systematic review of models for forecasting the number of emergency department visits. *Emergency Medicine Journal* 2009;26(6):395–399. doi:10.1136/emj.2008.062380
42. Hamilton JD. A new approach to the economic analysis of nonstationary time series and the business cycle. *Econometrica* 1989;57(2):357–384.
43. Höhle M, an der Heiden M. Bayesian nowcasting during the STEC O104:H4 outbreak in Germany, 2011. *Biometrics* 2014;70(4):993–1002. doi:10.1111/biom.12194
44. McGough SF, Johansson MA, Lipsitch M, Menzies NA. Nowcasting by Bayesian smoothing: a flexible, generalizable model for real-time epidemic tracking. *PLoS Computational Biology* 2020;16(4):e1007735. doi:10.1371/journal.pcbi.1007735

45. Salmon M, Schumacher D, Höhle M. Monitoring count time series in R: aberration detection in public health surveillance. *Journal of Statistical Software* 2016;70(10):1–35. doi:10.18637/jss.v070.i10